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OTHER ARTICLES

The Use of Botulinum Toxin for Nasal Esthetics: A Systematic Review

Zahra Sadat Modarresi[1] · Majid Beshkar[1] · Zeinab Bakhtiari[1]

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Abstract
Physical attractiveness is commonly evaluated based on facial appearance, with the nose being the most noticeable feature. Botulinum toxin injection is a less invasive method for enhancing nasal esthetics, which can address minor nasal imperfections through relaxing muscles. To fully understand the uses of botulinum toxin in the nose area, a thorough review of the available information is necessary. We aimed to conduct a systematic review of literature to evaluate the effects and clinical application of botulinum toxin for nasal esthetics. We searched PubMed, Scopus, Google Scholar, and the Cochrane Central Register of Controlled Trials (CENTRAL) databases up until May 2023 to identify studies investigating the use of botulinum toxin for cosmetic enhancement in the nose area. Data was extracted from the selected studies regarding the baseline characteristics, type of botulinum toxin used, dosage, injection site, and outcomes. Initially, 1139 articles were found and 23 articles were included. The studies showed the usage of botulinum toxin can correct drooping nasal tips, dilated nostrils, minor post-rhinoplasty deformities, and high columellar show. It can also augment nasal dorsum, improve cleft lip-related deformities, and address other disorders related to the function and appearance of the nose. Botulinum toxin injection seems to be a safe and effective treatment option for individuals seeking to correct minor nasal defects. Further large-scale studies are recommended.

Keywords Botulinum toxin · Nose · Non-invasive esthetic procedures · Nasal tip ptosis · Nasal flare · Systematic review

Introduction

Physical attractiveness is often judged by facial appearance, which has become increasingly emphasized in the modern world [1]. Enhancing the nose, the most prominent feature of the face, is a complex and fascinating procedure in the field of facial esthetics [2]. There are various treatment options available for enhancing the appearance of the nasal area, which range from non-invasive procedures to surgical interventions. Rhinoplasty is a frequently performed cosmetic procedure. However, patients often experience fear and psychological barriers when considering surgery [3]. It should be considered that undergoing rhinoplasty can lead to potential complications such as asymmetry, infection, and numbness. Moreover, a second surgery may be necessary in some cases [4].

Botulinum toxin (BTx) injection is one of the most popular used minimally-invasive cosmetic procedures with low complications. It can sometimes serve as an alternative to rhinoplasty [5, 6]. However, in cases of significant nasal deformities, complete improvement may not be achievable, and surgery remains the preferred option [7]. Botulinum works by relaxing muscles through preventing the release of acetylcholine [8–10]. Facial mimetic muscles are important in the overall and dynamic esthetics of the face. Their strong or asymmetrical forces can cause deviation and a decrease in the appearance of facial structures, including the nose [6, 11]. BTx injection can be a viable solution for minor nasal defects caused by anatomical issues or aging [12]. Indications for using BTx are diverse and include the drooping tip, alar flaring, and excessive columellar show [4, 12–14]. After the initial surgery, this technique can be utilized to address minor issues with the shape of the nose [12].

Previous studies were conducted on the safety or clinical application of BTx in general aesthetic treatments [15]

✉ Zeinab Bakhtiari
Bakhtiari_zeinab@yahoo.com

[1] Department of Oral and Maxillofacial Surgery, Dental School, Tehran University of Medical Sciences, Tehran, Iran

or cosmetic treatments of neck, and middle and lower face areas [16]. Also, global guidelines and consensus about the indications of BTx have been published [17, 18]. However, to the best of our knowledge, no systematic review has yet conducted about the use of BTx for nasal esthetics. Hence, this systematic review aimed to investigate the different uses of BTx in various types of nasal imperfections.

Methods

This systematic review was prepared based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [19]. A search to identify potential articles from the inception to May 20, 2023, was performed in PubMed, Scopus, Google Scholar, and the Cochrane Central Register of Controlled Trials (CENTRAL). We included studies on participants who used BTx injection for cosmetic improvement in the nose area. The control group, if present, was a group of patients who received a placebo or other standard cares. The desired outcomes included safety and efficacy measures which varied across studies. Those studies that used other interventions like dermal fillers, were not focused on nasal area were excluded. Also, in vitro and in vivo studies and book chapters were excluded. If the treatment protocol was examined in various regions of the face, but the nose area was assessed separately, the study was deemed qualified for inclusion. However, if treatment protocols in the same area, such as BTx and filler injections, were merged, the study was excluded.

The search terms used for this population, intervention, comparison, and outcome (PICO) strategy were ("rhinoplasty" OR "septorhinoplasty" OR "nose" OR "nasal") AND ("botulinum " OR "Meditoxin" OR "Botox" OR "Neuronox" OR "Oculinum" OR "Vistabex" OR "OnabotulinumtoxinA" OR "Onabotulinumtoxin A" OR "Vistabel"). Duplicate items were removed by EndNote software, and then two independent reviewers (Z.M. and Z.B.) reviewed the titles and abstracts of the articles according to the eligibility criteria. Then, the full texts of the articles were retrieved and reviewed for inclusion. They resolved disagreements through discussion and settlement.

Two independent reviewers extracted data from the identified articles. For each study, they extracted the baseline characteristics, the type BTx used, the dosage administered, the injection site, and the outcome measures. The discrepancies were resolved with discussion or consultation with the principal investigator.

Results

We found 1139 articles, including 285 articles from PubMed, 628 from Scopus, and 226 from Google Scholar. Also, 59 articles were found form CENTRAL. After removing duplicates, 957 articles were screened by title and abstract, and 39 articles were included for full-text review. The entire full text of these articles were reviewed and following the exclusion of 16 articles, finally 23 articles were included. Some of the reasons to exclude these articles were incomplete BTx data, use of both BTx and filler, not using BTx, and BTx usage only for rhytids (Fig. 1).

The included articles were categorized into three groups based on the type of the study. Findings of clinical studies, case reports/series, and viewpoints and expert consensus are presented in Tables 1 and 2, and Table 3, respectively. The mean ages ranged from 26.2 to 26.6 years in clinical studies (Table 1). The age range of participants of case reports or case series were from 19 years to 52 years (Table 2). Overall, the studies showed the usage of BTx can correct drooping nasal tips, dilated nostrils, minor post-rhinoplasty deformities, and high columellar show. It can also augment nasal dorsum, improve cleft lip-related deformities, and address other disorders related to the function and appearance of the nose.

BTx injections are effective in addressing the drooping of the nasal tip. In several studies, the dosage for these injections varied between 1.5 and 10 units, with 2 to 4 units emerging as the most frequently administered dosage. Also, administering BTx ranged up to 6 units in the dilator naris muscles on each side, with the most frequently employed dosage being 2–4 units at the center of the rim alar. Additionally, up to 5 units were used in the levator labii superioris alaeque nasi muscles on each side, with the most commonly applied dosage being 2–3 units, aiming to address excessive contractions of these muscles.

Discussion

Our findings showed the utilization of BTx for enhancing nasal esthetics. The applications of BTx for this purpose include correcting drooping nasal tip, dilated nostrils, post-rhinoplasty injection, and other disorders related to the function and appearance of the nose. The applications mentioned in the case studies are nasal dorsum augmentation, high columellar show reduction, and cleft lip repair. Further elaboration on these applications will be provided in the subsequent sections.

BTx injection is a minimally invasive option for correcting minor functional and aesthetic issues with the nose. Muscles such as the procerus, levator labii superioris

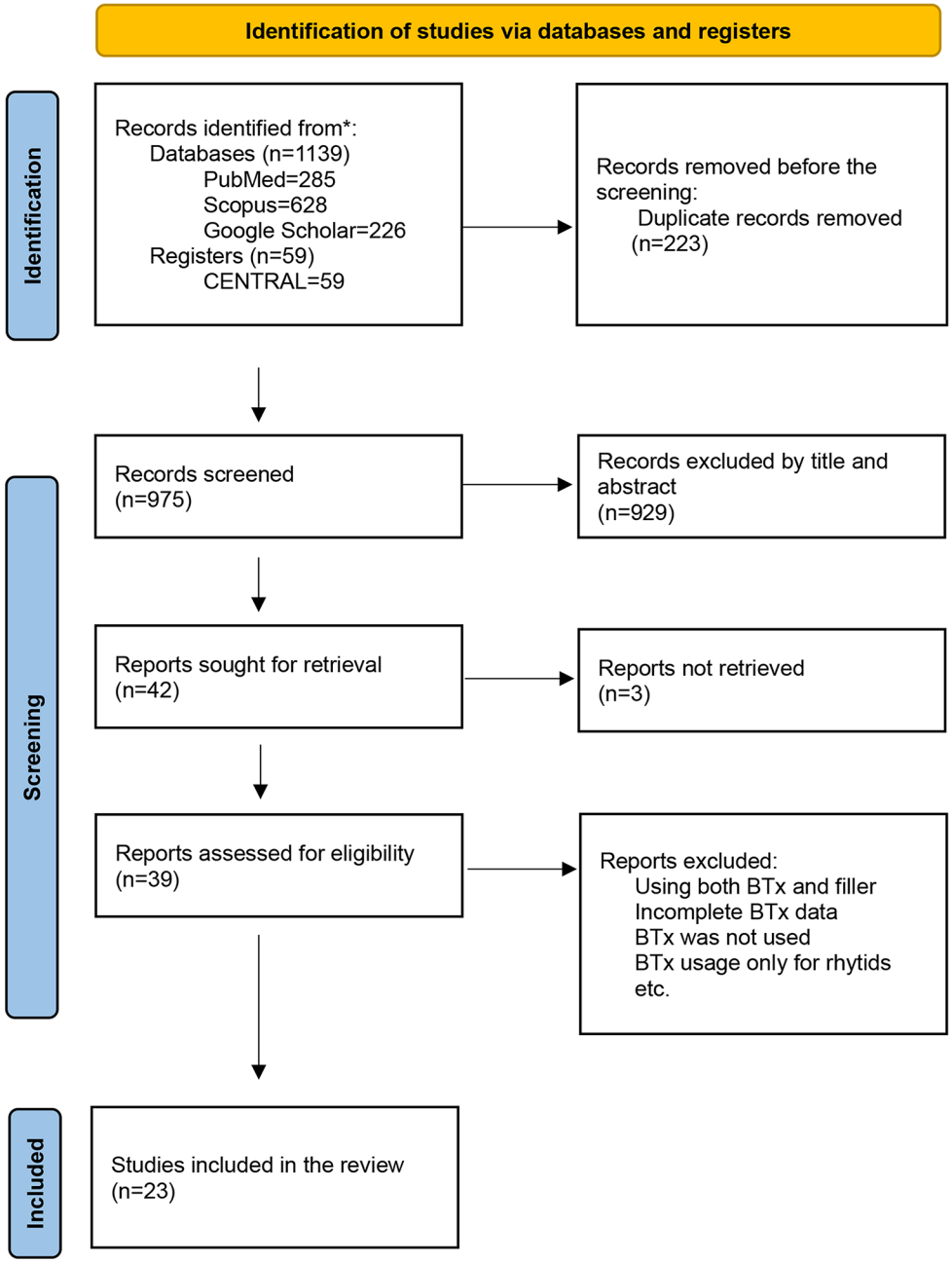


Fig. 1 Study selection process

alaeque nasi, dilator naris, depressor septi nasi, and nasalis can all affect the appearance of the nose [13]. Injecting BTx into these muscles can help alleviate certain issues related to the nose.

Rhinoplasty surgery, a commonly performed procedure, has a steep learning curve with many known complications [3]. like the recurrence of nasal deviations [11, 20]. Rhinoplasty has a longer recovery period and a higher price than using BTx to correct some nasal deformities [3]. BTx injection is a safe, effective, and less invasive method [8, 20]. Compared with surgery it has no wound, less downtime, less swelling, and almost immediate esthetic results [8]. It is crucial to have realistic expectations when contemplating BTx injections for the nose. While they may have a slight improvement in appearance, they are most suitable for patients with minor issues who do not expect significant changes. As a result, it is highly recommended to choose these methods in suitable cases. In patients where BTx injections are insufficient for nose reshaping, adjuvant methods such as filler injections can be used. During the

Table 1 Clinical studies that investigated the effect of botulinum toxin injection in improving the aesthetics of the nose

Author, year	Sample size	Age (year)	Botulinum toxin type & dose	Injection site	Outcome measurement	Follow-up duration
Redaelli [12]	43	42.7	Vistabex (Botulinum toxin type A, Allergan, Irvine, California, USA) -1.5 U per side	Depressor septi nasi just under the tip at the beginning of the columella	No longer drooping of the nasal tip while speaking (upturning of the nasal tip in some cases)	15 days
D'Emilio [32]	394 (12 cases of nasal area injection)	56.6	Onabotulinumtoxin A -2 U	The depressor nasalis	Only 3% of patients receiving full face and neck treatment needed BTx for Sagging nasal tip	NA
Lee [20]	60	Intervention: 32.3 Control: 30.8	Botulinum toxin type A (Botulax; Hugel Co., Chuncheon, South Korea) -4 U	Depressor septi nasi	-Esthetic evaluation using Adobe Photoshop CS6 (Adobe System, Inc) -Higher correction of deviation after corrective rhinoplasty in botulinum toxin–injected group -A significantly higher rate of excellent satisfaction in the experimental group	1 month postoperatively, and 6 months postoperatively
			2 U bilateral	Bilateral levator labii superioris alaeque nasi		
Salakshna [4]	14	35.8	-4 U Xeomin (Incobotulinum toxin; Merz Pharmaceuticals GmbH, Frankfurt, Germany) -7 U Dysport (Abobotulinum toxin A; Galderma Laboratories, Fort Worth, Texas)	Tiny aliquots on the midline and lateral portions along the upper nasal dorsum, targeted to the superficial layer of the procerus muscle	-Calculating volume change using the Quantificare camera -grading nasal augmentation subjectively by two blinded dermatologists using the photographs and by participants using a quartile grading scale -Overall nasal volume difference showing statistical significance (p-value of less than 0.001) using the Friedman test -Satisfaction with results for at least two months in all participants (consistent with the scores reported by the two blinded dermatologists)	Immediately after the BTx injection and at one week, two weeks, one month, and two months after the treatment
Zhong [6]	20	Intervention: 26.2 Control: 27.3	-3 U botulinum toxin in experimental group -Equivalent of normal saline in the control group	-The intersection of the dilator naris anterior, dilator naris vestibularis, and levator labii superioris alaeque nasi -The intersection of the dilator naris vestibularis, LLSAN, and dilator naris	-3dMDFace Dynamic surface imaging system records Standardized facial movement (from rest to maximum smile without revealing teeth). -The changes between rest and maximum smile statuses represented alar mobility and were generated By mobility=$\frac{WIDTHsmile-WIDTHrest}{WIDTHrest} \times 100\,\%$. -In the experimental group, alar flaring mobility decreased from 10.05%±6.40–4.91%± 3.48% ($P<0.05$), and alar base mobility decreased from 16.83%±5.69–12.50%± 4.89% ($P<0.05$). -Botulinum toxin type A can effectively restrain alar mobility and improve alar dynamic esthetic.	NA

Abbreviations NA: Not available; BTx: Botulinum toxin; U: Unit

pre-treatment examination sessions, the nose is evaluated both statically and dynamically. Patients should be mindful of any asymmetry in their appearance before treatment. It is recommended to take photographs of the nose in both resting and contracted states during each session [8].

Drooping Nasal tip

The depressor septi nasi muscle can cause drooping of the nasal tip if overactive [21]. This muscle originates from the orbicularis oris and periosteum superior to the central and lateral incisors, enters the septum, and depresses it as well as the posterior part of the alar [20–22]. The normal nasolabial angle in men is around 97 degree and in women is 104.9 degree [22]. The nasal tip can be raised with a BTx injection if the angle is less than 90 degrees [23]. Studies showed BTx injection can be used to treat sagging of the nasal tip. The injection dosage ranged from 1.5 to 10 units in various studies, with 2–4 units being the most commonly used dosage. The proper technique for administering the injection involves a vertical and deep delivery at the base of the columella, aligning with the nasal spine. This injection can

Table 2 Case reports and case series regarding the use of botulinum toxin for the appearance of the nose

Author Year	Age (year)	Patient problem	Treatment	Outcome
Dayan [33]	NA	Dynamic tip ptosis and cephalad pulled alar base	-5 U of Botulinum toxin type A (Allergan, Inc., Irvine, Calif) each depressor septi nasi -3 U of Botulinum toxin type A (Allergan, Inc., Irvine, Calif) each levator labii superioris alaeque nasi	-Less nasal tip ptosis and more neutral position of alar insertion while maximal smile -Increasing the nasolabial angle from 110 to 115 degrees
Aizenbud [31]	19	Hypernasal resonance and striking nasal grimace	-30 U of botulinum toxin type A (Dysport; Ipsen-Biotech Laboratories, Paris, France) each side -the zygomatic major and minor, levator anguli oris, and levator labii muscles	-Complete disappearance of the nasal grimaces -Impressive improvement in speech intelligibility -Significant reduction of nasal air emission -1 year postoperatively no nasal grimace detected
Turkmani [14]	40–52	The excessive columellar show (3 cases with upper columellar show)	-4 U of Anabotulinum toxin A each side -the levator ala nasi (medial portion of levator ala nasi)	-Remarkable reduction in upper columellar show in active and resting states -at a 4-month follow-up, the effect had almost vanished.
Eshraghi [30]	42 and 37	Synkinesis between the left procerus and orbicularis oculi muscles following cosmetic rhinoplasty (2 cases)	-Case 1: 8 U of Incobotulinum toxin A (Xeomin, Merz Pharmaceuticals GmbH, Frankfurt, Germany) -Case 2: 10 U of Abobotulinum toxin A (Dysport, Ipsen Limited, Slough, Berkshire, UK) -the procerus muscle (2 mm above the center of the horizontal plane that intersects both medial canthi)	The synkinesis improved and effectively resolved the Unintentional contractions.

Abbreviations UK: United Kingdom; U: Unit; NA: Not available

slightly raise the tip of the nose. It is recommended that the treatment be limited to patients with a hyperdynamic nasal tip which is aggravated during activities like smiling [8]. The immediate effects of injection are unremarkable. The duration of the effects ranges from 2 to 5 months in various studies [8, 24].

Pain is the most common complication reported in the treatment of nasal tip sagging by BTx injection [21]. BTx diffusion from the columella or injecting caudally and laterally can lead to dysfunction of the elevators of the upper lip and orbicularis oris, which causes asymmetry, flattening of the philtrum, and ptosis of the upper lip [24]. While rare, upper lip ptosis can occur when the depressor septi nasi is deeply paralyzed [21]. It is not advisable to administer injections to patients with elongated upper lips and lengthy caudal septum, as this may impede the desired rotation of the nasal tip [25, 26]. In patients with long upper lips and reduced vermilion show, caution is needed due to the high risk of lip ptosis [27]. These factors should be taken into account when considering BTx treatment for nasal tip sagging.

Nasal Flare

The nasalis muscle is shaped like a horseshoe, with its curved part made up of transverse fibers on the dorsum of the nose. The muscle's two lower sections are vertical and extend downward on either side of the nose [21]. These lower parts are known as the dilator naris and can contract during anger or discomfort, causing the nostrils to dilate [8]. The medial part of the levator labii superioris alaeque nasi muscle can also contribute to nostril dilation [18]. This muscle originates from the frontal process of the maxilla and enters the alar cartilage and the upper lip [20]. Hypertonicity of this muscle causes the nose to rise on the lateral side and the tip to rotate downward [23].

Our findings showed that up to 6 units of BTx injection in the dilator naris muscles per side with 2–4 units being the most commonly used dosage in the center of rim alar, and up to 5 units in the levator labii superioris alaeque nasi muscles each side with 2–3 units being the most commonly used dosage to treat excessive contractions of these muscles. In this regard, BTx injection in the dilator naris muscle can help with anatomical abnormalities not related to age but related to hyperactivity or pinched nostrils after rhinoplasty. This approach limits alar mobilities and relaxes the nostrils for 3–4 months [8]. Injections in the dilator naris muscles should be done with caution since there is a risk of dropping the corners of the mouth, ptosis of the upper lip, increasing the length of the philtrum, speech disorder and difficulty with eating and drinking with a straw [5].

Injection in levator labii superioris alaeque nasi muscles should also be done with caution due to the possibility of increasing the length of the upper lip. It is recommended that only experienced physicians administer this injection. This treatment is not recommended if the distance between

Table 3 The studies in which the authors recommend their viewpoints based on a review of studies or expert consensus about botulinum toxin usage in nasal aesthetics

Author Year	Problem	BTx type	Dose	Injection site	Outcome
Carruthers [34]	Repeated Nasal Flare		5–10 U bilaterally	The lower nasalis fibers draped over the lateral nasal ala	Decreasing of involuntary nostril flare in some patients
	Nasal Tip Droop		2–3 U	The base of the columella in Depressor Septae	Slightly elevating the nasal tip
Beer [13]	Static and dynamic nasal ptosis and nasal flaring	Botulinum toxin type A	2–6 U each side	Alar portion of the nasalis	Treating nasal flaring
			2–6 U	Depressor septi nasi	Raising the tip of the nose
			5 U each muscle (an additional 3 U injected into the levator labii superioris alaeque nasi on each side is also helpful)	Depressor septi nasi and levator labii superioris alaeque nasi	Ameliorating dynamic nasal tip ptosis
Ascher [21]	Drooping nasal tip	botulinum toxin type A (Speywood Unit)	10 s.U	The base of the columella	Slightly raising the nasal tip
Raspaldo [8]	Nasal tip repositioning dilated nostrils	Onabotulinumtoxin A	2–4 U	Deep injection perpendicular to the nasal spine	Minimizing tip drooping in patients with a hyperdynamic nasal tip
			1–2 U	Subcutaneous injection in the nostril dilator muscle	Relaxing nostrils
Redaelli [23]	Nasal tip rotation towards maxilla & nasal sides lift and downward rotation of tip	Vistabel/Bocouture Azzalure	2–4 U 10 U	Above the columella and in the nasal spine	Correction of the plunging nasal tip
		Vistabel/Bocouture Azzalure	2 U 10 U	The sides of the nose in levator labii alaeque nasi	
Gassia [35]	Sagging nasal tip Dilated nostrils	Onabotulinumtoxin A	2–4 U	Deep perpendicular injection at the level of the nasal spine in the depressor septi nasi muscle	Raising the nose tip position
			1–2 U per injection point	Subcutaneous Injection in the dilator naris muscle	Nasal flare narrowing
Ahn [5]	Nasal flare and depressed nasal tip	botulinum toxin type A	1 U per injection	Middle of each ala nasi	Nasal flare correction
			3 U	The base of the columella	Elevation of the nasal tip
Dormston [24]	Nasal tip ptosis and nasal flare	Onabotulinumtoxin A	3 U or 4 U or 8 U	Center of the columella	Nasal tip elevation
			1 U or 2 U	Lower end of the nasal wings	Nostril narrowing
Sundaram [18]	Nasal tip ptosis and nasal flare	Onabotulinumtoxin A	1–4 U	Dilator nasalis (alar portion of nasalis), Medial portion of levator labii superioris alaeque nasi may also be considered	Nasal flare correction
			2–6 U	Depressor septi nasi	Nasal tip elevation
Bertossi [26]	Sagging nasal tip	Onabotulinumtoxin A	2-4 U	The base of the columella	Pleasant upward rotation of the nasal tip and inhibition of its downward rotation when the patient speaks or smiles
Jones [22]	nasal tip ptosis and nasal flaring	AbobotulinumtoxinA	3 U (2–6 U for the treatment of hyperdynamic tip ptosis)	Columella, just above its junction with the cutaneous upper lip	Nasal tip elevation and treatment of hyperdynamic tip ptosis
			1–2 U	The alar portion of each nasalis	Decrease nasal flaring
Lapa [36]	Drooping Nose Nasal Flare	Botulinum toxin type A	1.5–6 U	Single midline injection into depressor septi nasi	Correct appearance of a drooping nasal tip during smiling
			4–10 U	The center of each alar rim	Inhibit widening of the nostrils seen with facial expressions

Table 3 (continued)

Author Year	Problem	BTx type	Dose	Injection site	Outcome
Signorini [25]	Sagging nasal tip	Onabotulinumtoxin	2–4 U	Full insertion of the needle into the base of the columella, directed at the nasal spine	Improvement of lowering of the nasal tip, especially during smiling
Wong [11]	Redeviation of the nose after septorhinoplasty	Botulinum Toxin (Botox 50 Allergan U: Allergan Aesthetics, AbbVie, Madison, WI)	7.5 U each side 2.5 U each side 5 U each side 2.5 U each side 2.5 U each side	Procerus Corrugator Nasalis Levator labii superioris alaeque nasi Depressor septi nasi	Blocking the pulling actions of the stronger or overacting nasal muscles in a crooked nose

Abbreviations U: Unit; BTx: Botulinum toxin

the nasal spine and the apex of the cupid is more than 1.8 cm [23].

Botulinum Toxin Injection after Rhinoplasty

Complications with the appearance and function of the nose may arise following rhinoplasty. BTx injections after surgery may help improve these conditions. One of these complications is the recurrence of pre-existing deviations. Also, new deviations in the shape of the nose may occur after rhinoplasty. The term "crooked nose" is defined as the deviation of nasal pyramids due to trauma, congenital, iatrogenic, or idiopathic causes [11]. By injecting BTx after septorhinoplasty, we can prevent a crooked nose. BTx injection can be used after rhinoplasty surgery to prevent nasal muscle movement and accelerate the muscle atrophy, allowing the nose to heal and stabilize in a straight position. Nevertheless, other studies are needed to prove this hypothesis [11].

Guarro et al. reported synkinesis as a possible complication after rhinoplasty [28]. Synkinesis is the involuntary contraction of facial muscles in response to voluntary contractions of other muscles. Some of the most frequently occurring forms of facial synkinesis are oral–ocular and ocular–oral synkinesis followed by ocular–nasal, ocular–chin, ocular–stapedial, chin–ocular, chin–oral, and platysma [29, 30]. Ocular–nasal synkinesis which is a contraction of the medial eyebrow muscles along with any deliberate or reflex blinking can occur as a rare complication several months after rhinoplasty [30].

In a study of two cases, contraction between the orbicularis oculi and procerus muscles produced ocular–nasal synkinesis within a few months after rhinoplasty surgery. BTx injection about 2 mm above the intercanthal line effectively eliminated these involuntary contractions [30].

Nasal Dorsum Augmentation

Salakshna et al. showed the ability to temporarily augment the dorsum of the nose with BTx injection. The study found that there was a significant increase in nasal volume immediately after treatment, and also one week after treatment. However, the increased volume after one month was not statistically significant. Nonetheless, the participants were satisfied with the treatment for two months. The injection was carried out in the midline and lateral portions of the upper nasal dorsum, targeting the superficial layer of the procerus muscle. By relaxing the outer layer of the procerus muscle, BTx can elevate the nasal dorsum [4]. This technique can serve as a temporary alternative to surgery or filler injections. When compared to other treatments, it tends to result in fewer and less severe side effects. However, the duration of its effects may be shorter. To accurately compare its effectiveness, further studies must be conducted.

Columellar Show Reduction

In a study by Turkmani et al., excessive columellar show was treated with BTx injections in each side of the levator ala nasi. Having an excessive display of the columella is not a genuine anatomical deformity. Instead, it is associated with the overactive function of the muscles. BTx can address upper columellar show, which refers to the medial alar portion of the levator labii superioris ala nasi. This muscle elevates the wing of the nose, resulting in a retracted alar rim. However, this procedure was only partially effective in treating combined shows. Due to its minimal impact on the depressor septi nasi, BTx is not recommended for addressing lower show, which occurs when the depressor septi nasi pulls the nasal septum down, making it more noticeable. After treatment, BTx had a considerable impact on all three patients in the span of three months. However, by the four-month follow-up, the effect had nearly disappeared [14]. These findings suggest using BTx as a temporary solution for excessive columellar show. However, further studies are required to recommend it as a predictable treatment method.

Cleft Lip-related deformities

Cleft patients with velopharyngeal inadequacy may experience nasal air emission that can lead to facial grimacing.

This behavior can distract listeners and contribute to negative stigma towards cleft patients. BTx injection can be used for temporary improvement. In a case report study, a boy was injected with 30 units of BTx in the paranasal muscles, including the zygomatic major and minor, levator anguli oris, and levator labii superioris alaeque nasi. This injection caused a significant improvement in speech and eliminated nasal grimaces [31].

Strength and Limitations

This is one of the pioneer studies that pooled findings of previous studies regarding the effects of BTx for nasal esthetics. However, it has several limitations that should be considered. Since we included different types of studies and due to the high heterogeneity, we could not perform meta-analysis. So, we used a narrative synthesis. Also, most of the included studies did not have a control group or compared the patients with placebo, so we cannot compare the outcomes of BTx injection with rhinoplasty or other surgical procedures. Moreover, we compared or reported the safety and efficacy outcomes of BTx injection for nasal esthetic. However, we did not evaluate the indications and guidelines for BTx injection for nasal esthetics. Future studies can consider this issue.

Conclusions

BTx injections in the nasal area can correct a variety of issues related to the appearance and function of the nose. For instance, the injections can be used to address drooping nasal tips, dilated nostrils, minor post-rhinoplasty deformities, high columellar show, and augment nasal dorsum. Additionally, BTx injections can improve cleft lip-related deformities and address other disorders that affect the function and appearance of the nose. Overall, BTx injection seems to be a safe and effective treatment option for individuals seeking to correct minor nasal defects. However, further large-scale studies are required to evaluate having realistic expectations is crucial.

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Declarations

Ethical Approval The manuscript has been read and approved by all the authors, the requirements for authorship have been met, and each author believes that the manuscript represents honest work. This article does not contain any studies with human participants or animals.

Informed Consent Not applicable.

Conflict of Interest None of the authors has any conflict of interest, financial or otherwise.

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